5(a)(i)	power = intensity × area	**C1**
	$= 1.3 \times 10^3 \times (\pi \times 0.055^2)$ $= 12$ W	**A1**
5(a)(ii)	intensity = power / area $= 12 / (\pi \times 0.0015^2)$ $= 1.7 \times 10^6$ W m^{-2}	**A1**
5(b)(i)	(λ =) v / f **or** c / f	**C1**
	(λ =) $3.0 \times 10^8 / 3.7 \times 10^{15} = 8.1 \times 10^{-8}$ (m)	**A1**
5(b)(ii)	ultraviolet	**A1**

Question	Answer	Marks
5(b)(iii)	$d \sin \theta = n\lambda$ **or** $(1 / N) \times \sin \theta = n\lambda$	**C1**
	$d = 1 / 2400 \times 10^{3}$ (m) $= 4.2 \times 10^{-7}$ (m) **or** $N = 2400 \times 10^{3}$ (m^{-1})	**C1**
	$n = 4.2 \times 10^{-7} \times \sin 90° / 8.1 \times 10^{-8}$ **or** $\sin 90° / 2400 \times 10^{3} \times 8.1 \times 10^{-8}$ $n = 5.2$ or 5.1 **or** when $n = 5$, $\theta = 76.4°$ and when $n = 6$, $\sin\theta > 1$ (so) $n = 5$	**B1**
	number of maxima = $(2 \times 5) + 1$ $= 11$	**A1**
5(b)(iv)	the wavelength has increased	**M1**
	(so) number of maxima decreases	**A1**